HW10

1. Proof: The covariance matrix $V(X)$ is positive & semi-definite.

$a^T V(X) a \geq 0$ for all $a \in R^n$

$$V(X) = E[(X - E(X))(X - E(X))^T] \quad (1)$$

A positive semi-definite matrix is a matrix whose eigenvalues are all non-negative, or equivalently:

M pos: semi-def $\Leftrightarrow$ $x^T M x \geq 0$ for all $x \in R^n$

For an arbitary column vector $a \in R^n$, we have

$$a^T V(X) a \overset{(1)}{=\!=} a^T E[(X - E(X)(X - E(X)^T) a \quad (2)$$

Since expected value is a linear operator, we rewrite (2)

$$a^T V(X) a = E[a^T (X - E(X)(X - E(X)^T a] \quad (3)$$

Define the scalar random variable

$$Y = a^T (X - \mu_X) \quad (4)$$

where $\mu_X = E(X)$ and note that

$$a^T (X - \mu_X) = (X - \mu_X)^T a$$

Combine (3) and (4), we have

$$a^T V(X) a = E(Y^2)$$

Since Y^2 is a random variable that can't become negative, and the expected value of a strictly non-negative random variable is also non-negative, we have

$$a^T V(X) a \geq 0 \text{ for any } a \in R^a.$$

HW 10.

2. Based on E.q (1.1),

$\|x\|^2 = x^T \cdot x$, then we have

$E(\|x\|^2) = E(x^T \cdot x)$

Claim 1.4 states $EX = 0$, claim 1.3 states $\mu = (\mu_1, \dots \mu_n)^T = EX$,

$\Rightarrow \mu = EX = 0$

Let A be the Identity matrix I, from claim 1.3 now we have

$E(X^T \cdot I \cdot X) = tr(I \cdot \Sigma) + 0$

$\Rightarrow E(X^T \cdot X) = tr(\Sigma) = tr(V(X)) = E(\|x\|^2)$

The trace of the covariance matrix V(X) is the sum of its diagonal elements, which are the variances of each component of x. They are equal to the eigenvalues $\lambda_1, \lambda_2, \dots \lambda_D$ of V(X).

Therefore, we proved $E(\|x\|^2) = \sum_{k=1}^{D} \lambda_k$.

From equation (1.10), we have $P_{V^*} = WW^T$, where $W = (V_1, \dots V_d)$ consist the first d eigenvectors of V(X).

The projection of X onto V^* is $P_{V^*} \cdot X = WW^T X$,

from E.q (1.1) we got $\|P_{V^*} \cdot X\|^2 = (WW^T x)^T (WW^T x)$

$E(\|P_{V^*} \cdot X\|^2) = E((WW^T x)^T (WW^T x)) \quad \mu = 0$

$= tr(WW^T \cdot V(X) WW^T)$

Trace Property: $tr(ABC) = tr(BCA) = tr(CAB)$

$E(\|P_{V^*} X\|^2) = tr(W^T V(X) W W^T W)$

$= tr(W^T W \Lambda^d) = tr(\Lambda^d)$

$= \sum_{k=1}^{d} \lambda_k$

Λ^d – diagonal matrix with eigenvalues $\lambda_1, \dots \lambda_d$ on it's diagonal.